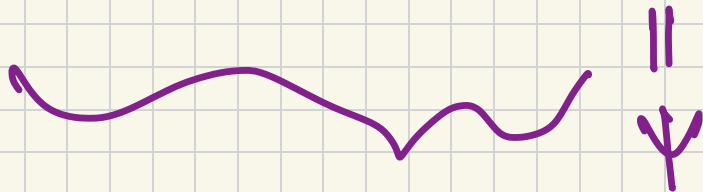


COL 703/7205

24th August

Derivations
in Hilbert's
proof
system

$\phi_1, \phi_2, \dots, \phi_n$



each ϕ_i is either
an instance of an axiom
or has been derived from
 ϕ_j and ϕ_k using
MP.

Deriver $p \rightarrow p$

1. $p \rightarrow ((p \rightarrow p) \rightarrow p) \rightarrow ((p \rightarrow (p \rightarrow p)) \rightarrow (p \rightarrow p))$

- instance of A2

2. $p \rightarrow ((p \rightarrow p) \rightarrow p)$

- instance of A1

3. $((p \rightarrow (p \rightarrow p)) \rightarrow (p \rightarrow p))$

MP 2, 1

4. $p \rightarrow (p \rightarrow p)$

- instance of A1

5. $p \rightarrow p$

MP 4, 3

We would like to prove that

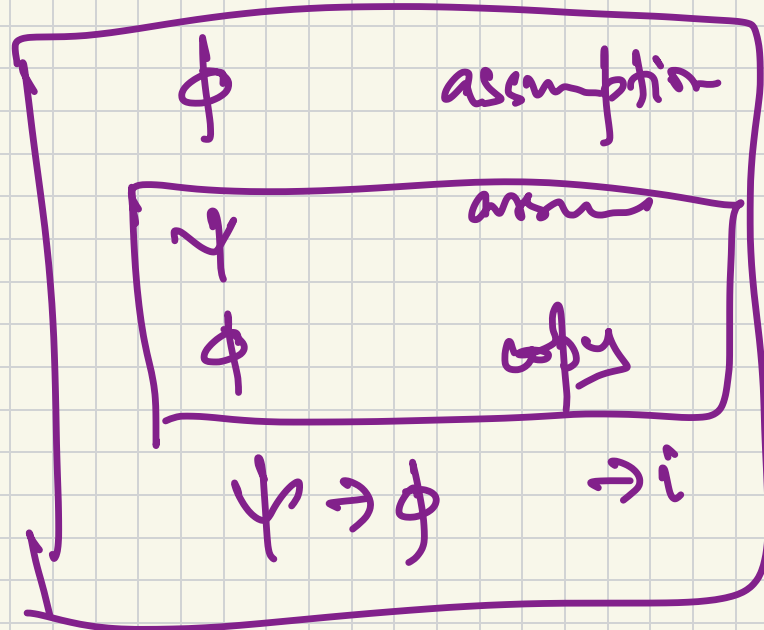
$$\vdash \phi \quad \text{iff} \quad \models \phi$$

Soundness If $\vdash \phi$ then $\models \phi$

Prove the axioms using
natural deduction.

(And then appeal to soundness of
natural deduction.)

$$\phi \rightarrow (\psi \rightarrow \phi)$$



$$\phi \rightarrow (\psi \rightarrow \phi) \rightarrow e$$

$$(\phi_1 \rightarrow (\phi_2 \rightarrow \phi_3)) \rightarrow ((\phi_1 \rightarrow \phi_2) \rightarrow (\phi_1 \rightarrow \phi_3))$$

$$\phi_1 \rightarrow (\phi_2 \rightarrow \phi_3)$$

assume

$$(\phi_1 \rightarrow \phi_2)$$

assume

$$\phi_1$$

assume

⋮

$$\phi_3$$

⋮

Exercise

$$(\neg \psi \rightarrow \neg \phi) \rightarrow ((\neg \psi \rightarrow \phi) \rightarrow \psi)$$

Soundness of
natural
deduction by
induction on the
length of the proof

(+ case by case
analysis of the
last rule applied)

Exercise: see
Huth & Ryan

ϕ_1
 ϕ_2
 $\vdots$
 ϕ_n
 $\vdash_1$
 $\vdash_2$
 $\vdots$
 $\vdash$

}

premise

if $\phi_1 \dots \phi_n \vdash \psi_i$
then
 $\phi_1, \dots, \phi_n \vdash \psi_i$

$\vdots$

rule

$\vdash_1 \psi_1 \vdash_2 \psi_2$

$\vdash_i$

Completeness

If $\models \phi$ then $\vdash \phi$

Every valid formula is derivable.

$(\neg \psi \rightarrow \neg \phi) \rightarrow (\phi \rightarrow \psi)$... TODO 1

We will prove the contrapositive.

If ϕ is not derivable then it is not valid.

Suppose we also prove that

$\neg \neg \phi \rightarrow \phi$

TODO 2

then if ϕ is not derivable, $\neg \neg \phi$ is also not derivable.

Consistency

We say that ψ is consistent
if $\nvdash \neg\psi$

($\neg\psi$ is not derivable).

Suppose we prove that
if ψ is consistent then ψ is
satisfiable **TODO 3**

If $\neg\neg\phi$ is not derivable then $\neg\phi$
is consistent and therefore $\neg\phi$ is satisfiable
 $\therefore \phi$ must not be valid.

TODOs

1.

$$(\neg \psi \rightarrow \neg \phi) \rightarrow (\phi \rightarrow \psi)$$

2.

$$\neg \neg \phi \rightarrow \phi$$

3.

If ϕ is consistent then
 ϕ is satisfiable.

We will do this in the
next class.

Restricted derivability

Let X be a set of formulas. We say that

ϕ is derivable from X

if

$\phi_1, \phi_2, \dots, \phi_n$

||

ϕ

where each ϕ_i is either an instance of an axiom or is coming from X or is derivable from X by MP on two formulas already derived.

Deduction Theorem

For a set of formulas X ,
and formulas ϕ and ψ

$$X \cup \{\phi\} \vdash \psi \quad \text{iff}$$

$$X \vdash \phi \rightarrow \psi$$

One direction is easy
if

$$X \vdash \phi \rightarrow \psi$$

applying MP

$$X \cup \{\phi\} \vdash \psi \leftarrow$$

$$\begin{array}{l} \vdash X \cup \{\phi\} \vdash \phi \rightarrow \psi \\ \vdash X \cup \{\phi\} \vdash \phi \end{array}$$

If $X \cup \{\phi\} \vdash \psi$ then
 $\underline{\underline{X \vdash \phi \rightarrow \psi}}$

Proof: by induction on the length of the derivation.

Base case : proof of length 1

$\swarrow$
 ψ is an instance of an axiom

$X \vdash \psi \quad X \vdash \psi \rightarrow (\phi \rightarrow \psi)$

$\searrow$
 ψ was present in $X \cup \{\phi\}$
 $\swarrow \searrow$
 $\psi \in X \quad \psi = \phi$

Inductive step



Next class